

Python Set Methods - Quick Reference

Most important Python set methods with descriptions and examples.

Method	Description	Example
<code>add()</code>	Adds a single element	<code>s.add(10)</code>
<code>update()</code>	Adds multiple elements	<code>s.update([20,30])</code>
<code>remove()</code>	Removes an element (KeyError if absent)	<code>s.remove(10)</code>
<code>discard()</code>	Removes an element (no error if absent)	<code>s.discard(10)</code>
<code>pop()</code>	Removes and returns an arbitrary element	<code>s.pop()</code>
<code>clear()</code>	Removes all elements	<code>s.clear()</code>
<code>copy()</code>	Returns a shallow copy	<code>new_s = s.copy()</code>
<code>union()</code>	Returns the union of two sets	<code>a.union(b)</code> or <code>a b</code>
<code>intersection()</code>	Returns common elements	<code>a.intersection(b)</code> or <code>a & b</code>
<code>difference()</code>	Elements in first set only	<code>a.difference(b)</code> or <code>a - b</code>
<code>symmetric_difference()</code>	Elements in either set but not both	<code>a.symmetric_difference(b)</code> or <code>a ^ b</code>
<code>issubset()</code>	Checks subset	<code>a.issubset(b)</code>
<code>issuperset()</code>	Checks superset	<code>a.issuperset(b)</code>
<code>isdisjoint()</code>	Checks for no common elements	<code>a.isdisjoint(b)</code>

Example Program

```
A = {1, 2, 3, 4}
B = {3, 4, 5, 6}

A.add(7)
A.update([8, 9])
A.remove(9)
A.discard(10)
print(A.union(B))
print(A.intersection(B))
print(A.difference(B))
print(A.symmetric_difference(B))
print({3, 4}.issubset(B))
print(B.issuperset({3, 4}))
print({1, 2}.isdisjoint({3, 4}))
C = A.copy()
print(A.pop())
C.clear()
```

`remove()` vs `discard()`

`remove(x)` raises `KeyError` if `x` is absent. `discard(x)` does nothing if `x` is absent.

Most Important Methods for Interviews

- `add()`
- `update()`

- `remove()`
- `discard()`
- `pop()`
- `clear()`
- `copy()`
- `union()`
- `intersection()`
- `difference()`
- `symmetric_difference()`
- `issubset()`
- `issuperset()`
- `isdisjoint()`